

Draftly

*A Lawyer-in-the-Loop Legal Workflow Platform for Sri Lankan
Legal Practice*

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1. Executive Summary

Draftly is a lawyer-in-the-loop platform for Sri Lankan legal workflows. The project has two main workflow tracks: conveyancing and litigation. The first prototype will validate the platform through conveyancing because title work is document-heavy, rule-driven, and suitable for measurable extraction, checking, and drafting. It reads a client's deeds and supporting records, extracts and links facts, reconstructs the title chain, identifies missing or conflicting evidence, and prepares editable outputs for lawyer review. The same platform direction later extends to litigation drafting and review, where pleadings and case documents can be organized around lawyer-approved facts, issues, rules, application, and conclusions.

2. Problem Statement

Sri Lankan legal drafting often begins with repeated manual reading, copying, and cross-checking of client documents. In conveyancing, lawyers and notarial clerks must identify the parties and property, trace ownership across prior deeds, check plans and extents, find missing instruments or encumbrances, and confirm that deed, survey, registry, and municipal details tally. In litigation, lawyers must organize facts, evidence, issues, authorities, and draft pleadings under tight procedural constraints. Source documents may be mixed Sinhala/English, scanned, photocopied, handwritten, tabular, or written in older legal styles. This makes both workflows slow and exposes them to transcription, consistency, and grounding errors.

Existing Sri Lankan platforms mainly support access to laws and cases. Draftly addresses the client-document workflow that precedes drafting in both conveyancing and litigation. General-purpose generative AI is not sufficient because an invented fact, citation, title link, pleading point, or authority can invalidate the output. Draftly therefore keeps each extracted fact, check, legal source, and draft traceable to source evidence and subject to lawyer approval.

3. Data Description

The project uses private matter data and reusable legal-source data. Private data will be organized by workflow. The conveyancing set consists of permission-based, anonymized bundles containing deeds, survey plans, registry extracts, municipal and assessment records, receipts, identity records, prior title reports, and lawyer-approved outputs. The later litigation set will cover pleadings, motions, affidavits, notices, evidence bundles, and lawyer-reviewed drafts when suitable data is available. Annotation will cover document type, parties, dates, references, extracted fields, and links between documents. Raw client names, identity numbers, addresses, signatures, and matter histories will not be published.

The reusable legal-source data currently includes:

Source layer	Count	Use in Draftly
Statutes	57	Rule and section lookup for workflow checks
Amendments	18	Tracking changes to statutory authority

Gazettes	7	Supporting delegated or official legal material
Conveyancing topics	20	Topic hierarchy for rule-based retrieval
CommonLII citation index	9,539 records	Broad case-law discovery layer
CommonLII matched judgments	3,703 judgments	Conveyancing-matched reported case layer
Official court records	5,474 records	Post-2012 judgment layer
Official court conveyancing matches	1,418 judgments	Conveyancing-matched official court layer

4. Methods

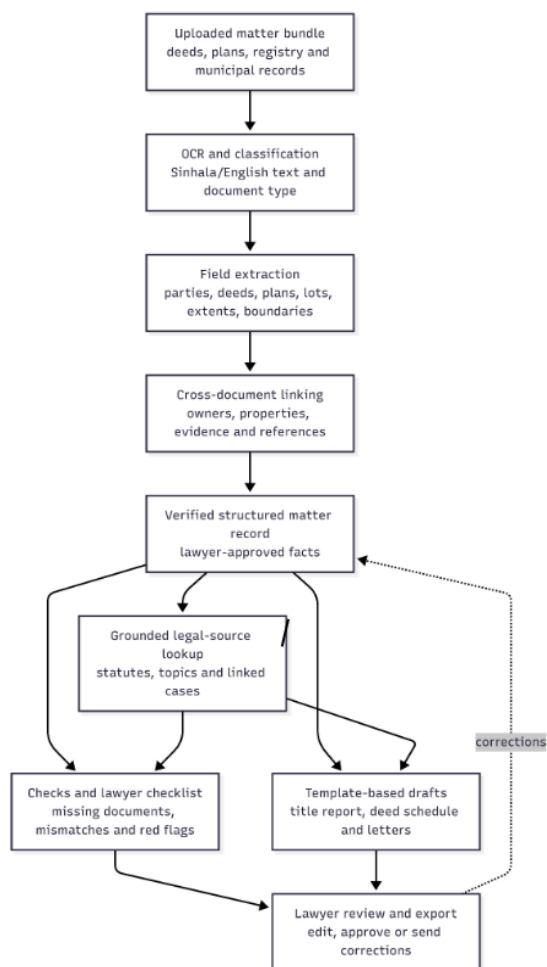


Figure 1. Draftly platform pipeline, shown through the first workflow: conveyancing.

The pipeline begins with document ingestion, OCR, and classification. OCR experiments will compare Surya OCR, Docling, cloud baselines, and layout-aware models such as LayoutLMv3 on selected Sinhala/English pages. The next stage extracts workflow-specific fields. For conveyancing this includes deed, party, date, plan, lot, extent, boundary, notary, registration, assessment, and receipt fields. For litigation this later becomes fact, issue, party, pleading, authority, evidence, and procedural-deadline extraction. Entity resolution and relation extraction then link names, properties, instruments, evidence, and references across the matter bundle to form the verified structured matter record.

Rule-based checks will first target conveyancing: missing deeds, inconsistent plan numbers or extents, unsupported transfers, encumbrances, identity and attestation issues, notary-jurisdiction concerns, and mismatches among deed, survey, registry, and municipal records. The litigation workflow will later use the

same pattern for issue lists, authority checks, pleading consistency, and missing evidence. Legal grounding will use deterministic retrieval over a BookRAG-inspired hierarchy: workflow step, topic, subtopic, statute section, and linked case. Curated keywords, citations, catchwords, and

explicit links will return source-backed material rather than generate legal authority. Approved templates will then produce drafts from lawyer-verified facts.

5. Evaluation Plan

Each critical component and the complete workflow will be evaluated, beginning with conveyancing. OCR will use character and word error rates together with field preservation. Classification will use accuracy; field and relation extraction will use precision, recall, F1, and normalized exact match. Cross-document linking will be compared with a lawyer-verified structured matter record using entity, relation, and title-chain agreement for conveyancing, with an equivalent issue-and-evidence structure for litigation in later work. Legal retrieval will use workflow questions with expected authorities and report recall at k, precision at k, and citation correctness. Drafts will be reviewed for factual errors, source traceability, lawyer-rated usefulness, and preparation time against the same anonymized matter completed manually. Any invented fact or legal authority counts as a failure.

6. Expected Outcomes and Success Criteria

The final project should demonstrate two lawyer-facing workflows: conveyancing and litigation. The first successful prototype should classify selected conveyancing documents, extract key fields, link facts into a verified structured matter record, display a title chain, identify missing or inconsistent evidence, retrieve grounded legal sources, generate a matter-specific lawyer checklist, and produce at least one title-analysis output and one editable conveyancing draft. The litigation workflow should be specified as the next platform track, with its data model, IRAC-style reasoning structure, and document templates prepared for follow-up implementation.

Success requires measurable title-chain agreement with lawyer-verified ground truth, correct and traceable legal-source retrieval, low factual-error counts, useful lawyer-reviewed drafts, reduced preparation time, and preserved privacy. The system remains assistive: the lawyer verifies, edits, approves, and exports every legal output.

7. Division of Work (Individual Responsibilities)

Lahiru – OCR and extraction: Benchmark OCR and document classification approaches; prepare document and field annotations; implement OCR, document classification, and field extraction pipelines; report OCR, classification, and extraction accuracy.

Himath - data and legal corpus: prepare the statutory and case-law corpus; build the topic index and deterministic retrieval engine; create and evaluate grounded legal-source queries.

Praveen – OCR and outputs: Benchmark and implement OCR approaches; define the output catalog; build approved parameterized templates and the shared deed schedule; implement draft generation and export; evaluate generated outputs for completeness, consistency, and factual traceability.

8. Preliminary Bibliography

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